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Chapter 3: Research Methodology and System Design

3.1 Introduction

In this chapter two parts are provided, one methodological and the other interrelated in objectives: the former gives the methodological underpinning of the study; the latter is the analysis of the objectives. The first part outlines the systematic literature review (SLR) process as detailed in the PRISMA 2020 guideline (Page et al., 2021) which led to initial pool of 70 studies reduced to 32 in Chapter 2, and is then followed by a description of the second part, namely the scoping review (SR) process, further reducing the studies to 25. The first part describes the process of conducting a systematic literature review (SLR) in accordance with the PRISMA 2020 guideline (Page et al., 2021) which reduced the initial pool of 70 studies to the 32 studies synthesised in Chapter 2, followed by description of the second part, the scoping review (SR) process, which further reduced the studies to 25. The second part takes the gaps identified in that review and translates them to a conceptual system design showing the data flow and workflow architecture in which the system's algorithmic components, explainability, and clinical integration would function to tackle the data challenges they present. Like typical systematic reviews, the second part of the chapter contains a description of the design decision and leads from the first part to the methodological or evidential finding which provided justification. As in a typical systematic review, the second part of the chapter outlines the design decision and traces back to the first part to the corresponding methodological or evidential finding that justified the design decision.

3.2 Research Design and Philosophical Positioning

The method is an unusual blend, with two quite different traditions in the study: One tradition is the interpretative and comparative approach, with which the study is performed, and which deals with secondary sources; the second tradition is the constructive approach to system design, in which the study is seen as an artefact production. The outer layers show the methodology used in the study, as this was drawn from a broadly pragmatist philosophical positioning whereby the "fitness for purpose" of the method is being considered in terms of its contribution to answering the research question, as opposed to "fostering a particular position in relation to ontological assumptions and the structure of knowledge". Saunders et al (2019) which underpins this work. A purist's orientation deliberately: if it was purely positivist namely, that only the empirical and objective study of (a body of) facts is to be examined but

not their design consequences, then it wouldn't work so well with the system-design component which asks the researcher to make normative considerations about how gaps in an existing body of empirical evidence should be expressed in design features. A strict interpretivist perspective, on the other hand, would be limiting the structured, criterion-based and auditable character of the review process used to implement PRISMA. A pragmatist stance enables both components to be implemented in a rigorous manner where they will retain their own individual research designs but within the same structure.

From a research strategy perspective, the study has a two-step design, with the evidence-synthesis component design with an SLR strategy and system-design component design with design science research (DSR) approach following the process model of Peffers et al. (2007) and Hevner et al. (2004). The pairing is methodologically sound because DSR itself explicitly relies on the premise that a design artefact's specification should be based on a sound knowledge basis in this study the SLR fills both the knowledge bases of a strong theory (the identified and discussed research gap) and a strong technical knowledge (the algorithmic and explainability precedents documented in Chapter 2) for the design artefact that is to be proposed. The time horizon is cross sectional (as to SLR), because it looks at how the literature stood at one point in time, and iterative (as to design) because the DSR process model used to design artefacts by Peffers et al. (2007) is explicitly non-linear and allows the researcher to revisit past design decisions as new constraints arise in the further stages of artefacts specification.

3.3 Systematic Literature Review Methodology - PRISMA Framework

3.3.1 Rationale for Using PRISMA

We implemented the PRISMA 2020 declaration (Page et al., 2021) as the reporting and process framework for this literature review in recognition of the importance to ensure that literature reports on the evidence base underlying AI therapeutic tools are transparent, replicable and auditable, a weakness documented in many of the AI-diabetes related literature reviews themselves (Masood et al., 2026; Mohsen et al., 2023). PRISMA was developed as an instrument to facilitate reviews of the effectiveness of healthcare interventions (Moher et al., 2009) and has since been updated to stay in line with methodological advances in reviews (Page et al., 2021); it is not only applicable to intervention effectiveness reviews but also to other review types, such as prevalence, prognosis and, in the present case, reviews of the technical and methodological features of a set of prediction-modelling studies. While PRISMA was born

from clinical and health-sciences research, its approach to search, screening and selection is equally relevant to the literature used to review the computational and methodological literature consolidates in Chapter 2 and its use is further justified by its application to reviewing other types of literature review (with a technical or scoping character) that has been reported in PRISMA style (Tricco et al., 2018).

3.3.2 Research Questions Guiding the Review

What machine learning and deep learning methods have been used in diabetes risk prediction, and what have been reported as their performance, validation characteristics? (RQ1) (RQ2) What is the evidence of the application of these within Ghana and across Africa? (RQ3) How much is clinical decision support used in diabetes prediction research, and what is the current gap in using both clinical decision support and explainable AI techniques? Throughout the search strategy below, and data extraction coding, these questions were used to guide.

3.3.3 Search Strategy

To search the literature, four databases were chosen (PubMed/MEDLINE, Scopus, IEEE Xplore, Google Scholar), having disciplinary breadth both in the computational literature and in the clinical literature related to the review. This combination is similar to earlier computer science and health sciences oriented systematic reviews of neighbouring areas which typically relied on merging data in one database for sufficiently high recall numbers from both fields (Kitchenham et al., 2010). To guarantee a sufficient amount of African related studies are retrieved, as observed in Chapter 2, that African-context studies are relatively underrepresented in the mainstream indexing services, a supplementary geographic filter term ("Ghana" OR "Africa" OR "Nigeria" OR "Ethiopia" OR "sub-Saharan") was applied during the second wave of searching. Search strings were then structured into three concept clusters: (i) the condition of interest diabetes or type 2 diabetes or diabetes mellitus, (ii) the technical approach machine learning or deep learning or artificial intelligence or explainable AI or XAI, and (iii) application context prediction or risk prediction or clinical decision support or CDSS.

Due to the fast publication pace of applied machine learning research in general, the search was restricted to studies published during the period 2015-2026, when SHAP- and LIME-based explainability methods become popular after Ribeiro, Singh & Guestrin (2016) and Lundberg & Lee (2017), and to English-language peer-reviewed journal articles, conference proceedings,

and, in case no peer-reviewed version is available, high-quality preprints (e.g., Adda, 2025; Khokhar et al., 2025).

3.3.4 Inclusion and Exclusion Criteria

Screening eligibility was evaluated on well-defined inclusion and exclusion criteria across all stages of screening and applied uniformly. Studies included those that: (a) used machine learning, deep learning or explainable AI techniques to predict, classify and support clinical decisions around diabetes; (b) provided empirical findings based on a specific dataset; and (c) were published in English in the timeframe defined. The studies that were excluded were those that: (a) concerned diabetes management or treatment, instead of risk prediction or the diagnosis of diabetes; (b) addressed only gestational diabetes or Type 1 diabetes, and did not appear to contribute in a way that could be generalised to Type 2 diabetes, despite explaining, or being part of, a CDSS-architecture; (c) were either narrative commentaries, editorials, or opinion pieces (as these are not based on empirical evidence) or; (d) did not report enough details (algorithm, dataset, and evaluation metric) to extract the data. These ones are similar to the Kitchenham/Charters (2007) general practice of having a trial and error with candidate papers using a small sample before conducting the screening process' and here they were tried with a small number of papers (ten) in this instance to ensure they were consistently applied.

3.3.5 Study Screening and Selection Process

Screening was conducted in three phases following the PRISMA 2020 for screening (Page et al., 2021). First, the titles and abstracts of all the records retrieved from the four databases were assessed for possible inclusion and/or exclusion according to the inclusion and exclusion criteria; clearly irrelevant records, such as studies concerned with glycaemic control algorithms for insulin dosing, diabetic retinopathy image classification, were discarded at this stage. Second, the remaining documents were retrieved and judged on the same criteria with reasons explaining why any studies were excluded for each document to comply with PRISMA guidelines for transparency. Third, the included Ref Lists of all the studies that passed the full text screening were hand-searched for additional eligible studies that were not identified by the screening searches in the databases—a supplementary method sometimes referred to as backward snowballing” and beneficial for identifying those studies with regional African studies that may be inconsistently indexed (Ukoba et al., 2025). After duplicate records were identified in the four databases (previously removed, see above) through reference management software, the titles/abstracts were removed. Prior to screening the titles/abstracts,

all duplicate records identified in the four databases were removed using reference-management software.

3.3.6 PRISMA Flow Diagram

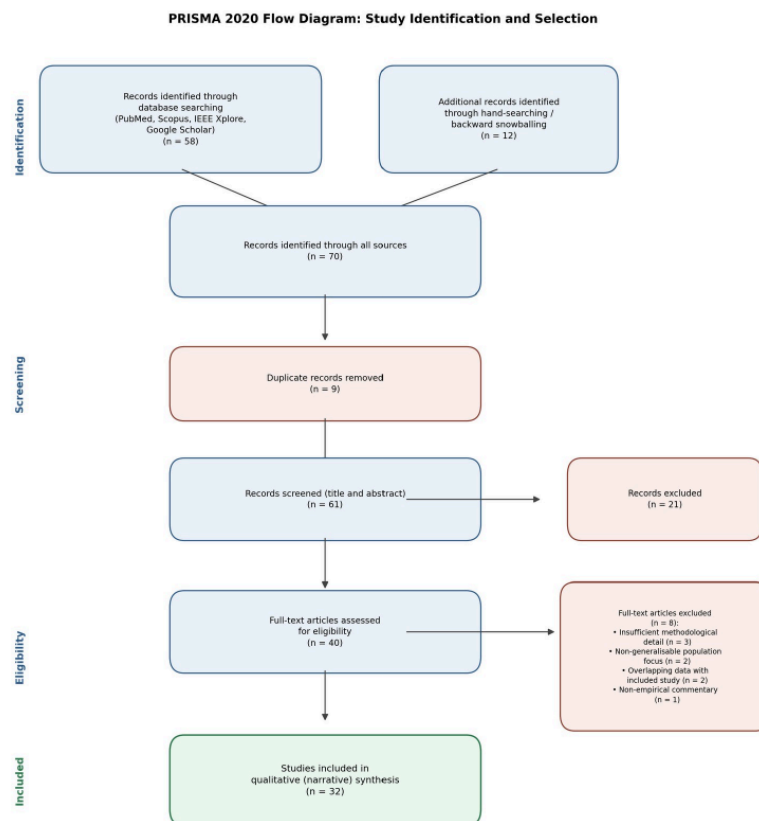


Figure 1 PRISMA 2020 flow diagram showing the identification, screening, and inclusion of studies (n = 32).

A checklist of the PRISMA elements is used to summarise the review process in the following way, with a standard PRISMA flow diagram structure. A total of 70 were found in the database and some of these were deleted as duplicates before screening. The rest of the records were filtered at the title-abstract level, and many were eliminated for being obviously topical irrelevant (for instance, articles related to diabetes complications, insulin pump engineering and prediction of non-diabetes chronic diseases). The rest of the records went through full-text evaluation, where additional exclusions were applied for low-quality reporting of methods, select clinical subgroups that are not applicable to the general public, or when data was repetitive and could have been covered by a more comprehensive study by the same author

group. The final set of 32 studies was a staged process which resulted in the review and synthesis of studies presented in Chapter 2. The presentation of this process in the form of a flow diagram and not solely as prose would now form a key component of a PRISMA reporting requirement (Page et al., 2021) for making the comprehensiveness of the review easily verifiable by those other than the original review team, while simultaneously making it harder for the review team to conceal how many of the literature they had found failed to meet the study's evidentiary criteria.

3.3.7 Quality Appraisal

Considering the types of study designs captured in the review that ranged from retrospective cohort analyses, algorithm-benchmarking studies using public datasets, to prospective clinical validation studies, a single risk-of-bias instrument intended for one particular study type (e.g., a tool only intended for RCTs) was deemed unsuitable. The Mixed Methods Appraisal Tool (MMAT) (Hong et al., 2018) was chosen as this tool includes five parallel sets of quality criteria that have been developed to evaluate qualitative, randomised, non-randomised, quantitative descriptive and mixed methods designs, which means that individual studies could be appraised against the quality criteria relevant to their methodological approach without compromising an overall appraisal framework. All of the 32 studies in the analysis were evaluated against the two screening questions used for the MMAT (clarity of research questions; adequacy of collected data to address the research questions) and the 5 criteria specific to each design category, while paying special attention to sample representativeness, appropriateness of the outcome measure, and completeness of reported results criteria that match the weaknesses found analytically in Chapter 2 (Mohsen et al., 2023; Masood et al., 2026). Further, and particularly for the predictive studies that were included, the appraisal was guided by the appraising domains outlined in the reporting sections of the TRIPOD statement (Collins et al., 2015), even though this was just a reporting guideline, and did not provide a risk-of-bias tool, it was still a useful additional checklist for assessing the extent to which a study had the results from its model development, validation, and model performance reported further in sufficient detail to allow for conclusions to be drawn from it in Chapter 2, where as stated in Section 2.2, the studies reviewed often lacked a full report in this domain of the appraisal.

3.3.8 Data Extraction Procedure

Details regarding the 32 included studies regarding its: bibliographic information; geographic/population context; algorithm(s) evaluated; characteristics of the dataset (source,

size, and reporting of class-imbalance correction), if any; its performance metrics (accuracy, AUC, sensitivity, specificity, and, if reported, calibration statistics); application of an explainability method (SHAP, LIME, or another technique); integration of or intention to integrate with a CDSS architecture; limitation statements, if any, that were explicitly formulated. This extraction framework applies to the four strands of review that were created in Chapter 2, and allowed for the subsequent narrative synthesis (Section 3.5) to be conducted in a systematic fashion, not impressionistic.

3.4 Characteristics of Included Studies

The 32 studies included ranged between the year 2005 (for foundational CDSS-effectiveness studies, Garg et al., 2005 and Kawamoto et al., 2005) and 2026, spanning the review that included both CDSS-effectiveness studies (foundational) and the most recent diabetes-specific CDSS studies utilized for XAI. Studies included in this review map to three regions: General or high income-context ML/DL studies with no particular region of interest (e.g. Choi et al., 2020; Rhee et al., 2021; Muhammad, Algehyne and XAJ, 2020); African-context studies with a Ghana, Nigeria, and Ethiopia focus (e.g. Adua et al., 2021; Dada, Birma and Gora, 2024; Anteneh et al., 2025); and studies focused on CDSS or XAI methodology with no geographic focus (e.g. Sutton et al., 2020; Abbas, Jeong and Lee, 2025). Algorithmically, tree-based ensemble methods (random forest, gradient boosting, XGBoost), and logistic regression are predominant, with few studies describing deep learning architecture, but primarily those using very large, nationally representative datasets (e.g., in the absence of other algorithms, such as in Rhee et al. (2021); Kelkay et al. (2026)). A small number of studies in fact a relatively small sample explicitly incorporate an explainability method (Chari et al., 2023; Rafie et al., 2025; Khokhar et al, 2025; Maimaitijiang, Aihaiti and Mamatjan, 2025; Majyambere et al., 2026; Malluhi et al., 2026) and even fewer connect explainability with a formal CDSS architecture (Du et al., 2022), which further supports the research gap identified in Chapter 2.

3.5 Synthesis Method

As there was significant variation in study design, outcome definition and metrics reported across the studies (as shown in Section 3.4), a quantitative meta-analysis was not considered suitable for this review (revealing a broader reporting trend in AI-diabetes as highlighted by Masood et al. 2026). Combining the results of studies that use different algorithms, use different datasets, and, in some cases, actually try to solve different underlying predictive tasks

(in fact, which should be noted prospective onset prediction vs. cross section classification, as described in Section 2.1 below) will likely result in an accuracy or AUC that is numerically accurate but very conceptually meaningless. Rather, the authors adopted a narrative synthesis approach for its general justification presented in the tertiary (review) systematic-review papers of the software and information-systems research community in HUDSS that the synthesis should take narrative form when the included primary (research) studies are methodologically heterogeneous (Kitchenham et al., 2010). The narrative synthesis was organized by the four review strands listed in Chapter 2; each of these was separately coded from the data extracted from records for each of the rows in the data-extraction matrix (see Section 3.3.8); the cross-strand synthesis organized by these four strands is presented in Section 2.6.

3.6 From Evidence Synthesis to System Design

The reasoning approach from evidence synthesis to creation of the design artefact in this study follows that of Hevner et al. (2004) in design science research, which is that the design artefact is necessary to solve a problem, the importance of which, and the presence of a current lack of adequate solution for which, is justified and based on a knowledge base. The knowledge base for the artefact in hand is directly responsible to the synthesis given in Chapter 2, and their relation to each other is not an incidental relation. The lack of externally validated and calibration-aware predictive models with Ghanaian data that were found in the literature (Section 2.2, 2.3) will guide the approach for developing models (Section 3.10); the lack of studies finding application-grounded and patient-specific evaluation of explainability (Section 2.5) grounds the explainability integration point (Section 3.9) in the workflow design; the absence of any addressed study that focused on low-infrastructure deployment (Section 2.4) will inform the data-flow architecture (Section 3.8); and the lack of any addressed data-governance framework (Section 2.6) will inform the ethical considerations (Section 3.12). This traceability is reflected in the specification of the artefact presented in the remainder of the chapter, in line with Peffers et al., (2007) who recommended that a statement about a design science artefact should explicitly and clearly trace back to its problem identification.

3.7 Conceptual System Design

3.7.1 System Overview and Design Objectives

The proposed system aims to be a clinician-facing explainable CDSS to complement, not supplant, clinical decisions about diabetes risk in the context of Ghanaian primary and

secondary cares. The three design objectives are drawn out of the literature synthesis and they address the key constraint cited in the literature regarding transferability of the models (Section 2.3), the locally-versus-global explainability distinction (Section 2.5), and the digital infrastructure maturity constraint (Section 2.4).

3.7.2 Conceptual Architecture

Four logical levels make up the conceptual architecture. Patient-level clinical and demographic input is provided to the data layer either by direct input by the clinician or (if available) extracted from an existing EHR. The model layer contains the defined predictive algorithm (or in the development mode, the set of candidate algorithms being compared, as outlined in Section 3.10) which can make risk inferences on the processed input. The explainability layer produces an explanation that translates the model's performance back to an individual patient, based on a feature-attribution method similar to the one used by Du et al. (2022) and many other works in diabetes-XAI reviewed in Section 2.5. The interface layer shows the prediction and accompanying explanation to the clinician, and does so in a manner which promotes speed, and therefore low cognitive burden, in interpreting the prediction when applied to clinicians' time concern raised in Section 2.5 for interpreting their prediction at the point of care (Sutton et al., 2020). This is a four-layer separation, based on a general architectural principle that was developed in CDSS literature that decision-support functionality should be modular with respect to its data, inference and presentation elements, such that any one layer (e.g. the underlying algorithm) could be redesigned without requiring redesign of the others (Berner 2009; Wright et al. 2009).

3.7.3 Design Principles Derived from the Literature

The conceptual architecture is found in four design principles that can be traced to specific weaknesses found in Chapter 2. To complement this criticism (presented in Section 2.4) on explainability becoming a post-hoc analytical process, interpretability-by-design puts the explainability layer at the heart of the inference pipeline, instead of tackling it on as an afterthought. The data layer needs to accommodate automated capture of data from the EHRs and manual clinician entry (adapting to infrastructure requirements, as described in Section 2.4 Wright et al., 2009). In order to overcome the discussed risk factor, disconnect in this section (2.3), locally grounded feature specification takes into account epidemiological evidence from Ghana and as a result the input features of the model layer are selected based on the local, rather than relying on international benchmark datasets (Gatimu, Milimo and Sebastian, 2016). To

perform the calibration-aware evaluation, the model layer should not only be assessed with respect to its discriminative performance but also take into account the agreement between the predicted and measured risk levels, given that we lack calibration reporting in current deep learning models (Mohsen et al., 2023).

3.8 Data Flow Design

3.8.1 Data Sources and Inputs

The system's data inputs are organized into three groups based on the evidence for each epidemiological risk factor of the disease reviewed in Chapter 2: demographic variables, physical activity level and dietary pattern, where available, and family history of diabetes are clustered under; blood pressure, body mass index and, where available, fasting plasma glucose or HbA1c are clustered under clinical and anthropometric variables.

3.8.2 Data Flow Diagram

The data flow architecture is shown with the levelled data-flow-diagram (DFD) convention which was originally formalised by DeMarco (1979) and which is well suited to showing the movement of data between the system's data-capture, processing, and presentation layers, without referring to the control logic specifying when the various transformations occur. At Level 0 (Context Diagram Level), the system is represented as one process, with three external processes that supply it with patients' data, and to which it returns risk prediction and explanation – the clinician, the local/central patient data store (either already existing or a stand-alone version or manual store with patient data, where there is none), and finally, the model training environment (where it receives trained and validated version of the model periodically for later deployment into the live environment). At level-1 it is broken down into 4 constituent processes corresponding to the architectural layers described in section 3.7.2: (P1) Data Capture and Validation: the aetiology of the patient's data is clinical or EHR based; completeness checks, range checks, etc. are performed on data before it is forwarded to the next layer; (P2) Preprocessing and Feature Construction: feature-construction transformations apply; (P3) Risk Inference: application of the learned model to the pre-processed feature vector to generate a probability estimate; and (P4) Explanation Generation and Presentation: output of explanation for risk-inference process (P3): output from P3 is local to the feature and Human Explanation generation processes respectively which perform a patient-specific feature-attribution explanation to the risk-inference process output and preprocesses the risk-inference

process output and a feature-attribution explanation along with the predicted output and transmits them to P4, the explanation-generation Process for display for the Clinician. Data is stored between each process (in a validated-input store between process P1 and P2, between process P2 and P3, and in a processed-feature store); this is required for auditability, and in infrastructure settings will enable the re-calibration of the model at a later date.

3.8.3 Data Preprocessing and Feature Engineering Pipeline

The following four functions are performed by the preprocessing pipeline: handling the missing values in the data (to address the concern about data incompleteness outlined regarding single-clinic African datasets in Section 2.3); encoding of other non-numeric clinical variables (to address the concern about non-numeric variables discussed in Section 2.3); class-imbalance correction (to address the class-imbalance weakness discussed in Section 2.2); and appropriate feature scaling for the family of candidate algorithms being evaluated (to address the support of the feature concern discussed in Section 2.1). Like the rest of the CRISP-DM pipeline this one is not a standalone preparatory process but a step which may be done again in the modelling stage (Section 3.10) when feature-engineering requirements are discovered which were not originally specified.

3.9 System Workflow Design

3.9.1 End-to-End Workflow

The proposed process would start with the clinician asking the patient to complete an either opportunistic or routine consultation where a risk assessment is completed, manually entered data and/or retrieved from the existing EHR (if available). The system follows the sequence of data flows outlined in section 3.8.2, and finally generates a risk score prediction and its explanation. The clinician then examines the prediction/explanation and either agrees, challenges or rejects the system's output and records the decision, in accordance with his clinical judgment. This sequence explicitly safeguards the clinician's authority over the end decision in response to clinical concerns that an unclear or overly standardized output from AI may excessively boost clinicians' confidence (Sutton et al, 2020).

3.9.2 Workflow and Decision Points

To operationalise the design principles outlined in section 3.7.3, there are two points in the workflow where a decision needs to be made. The first decision point is typically at the same

time as data is captured, since the system checks for the availability of sufficient structured data to move forward to automated inferring; if key structured data elements are missing, then at this point, the system asks the clinician to manually provide the missing value to feed into the subsequent automated inference, or alternatively, skip to performing this reduced-confidence prediction with an explicit warning for the clinician, not to have the value automatically substituted into the analysis in a way that could lead the clinician to believe that the prediction has high confidence when in fact it only has reduced confidence. After the clinician has read the prediction/explanation, the second decision point is when the Clinician writes their accept/override decision, and when possible, a brief note describing the reason for the reject or accept of the clinician, is made. The second decision point is specifically aimed at collecting the process-level usability data that is lacking in the current CDSS-effectiveness literature (Kawamoto et al., 2005; Garg et al., 2005), and thus gives the evaluation approach described in Section 3.11 an evaluation mechanism that involves more than just model-level performance measures.

3.9.3 Explainability Integration Point

The output of explainability comes from the risk prediction itself and is integrated into the workflow without being a separate risk prediction analysis screen or an added option that the clinician could select – following the principle established analytically in Section 2.5 that an explanation needs to be presented at the point of decision, with no need for an additional clinician-initiated request. The meaning is conveyed to the reader as a ranked list of the parent-specific features that play the most important role in the predicted risk score (a distinction that we identified as a gap in the literature on applied XAI discussed in Chapter 2 between local and global explanations), as opposed to a global summary of the feature importance.

3.10 Model Development Approach

The study's methodology in developing the model, as outlined in the modelling phase of the CRISP-DM process (Chapman et al., 2000), is explicit rather than dominionistic, meaning it is not predetermined that it will be based on just one algorithm, but it explicitly takes place in a comparative way, in direct response to the inconsistent and non-comparable performance reporting that has been found across the studies undertaken for the African context that are reviewed in Section 2.3 (Ukoba et al., 2025). It is noted at the end of Section 2.2 that deep-learning alternatives do not prove to be the standard option to select in favour of structured, tabular clinical data where the algorithms of choice are from the ensemble and tree families.

Explainability is achieved through SHAP (SHapley Additive Perturbations) (Lundberg and Lee, 2017), chosen over LIME (Local Interpretable Model Agnostic) (Ribeiro, Singh and Guestrin, 2016) due to stronger theoretical consistency guarantees while keeping up the drawback in terms of computational cost, which is addressed at architectural level with the modular layer of explainability described in Section 3.7.2.

3.11 Evaluation Strategy

An evaluation strategy has been prepared that is designed to be more inclusive than discrimination alone, as has been censured in section 2.2 and 2.4. Evaluation is measured by the reporting domains of the TRIPOD statement (Collins et al., 2015), which include the discrimination metrics commonly reported (AUC, sensitivity and specificity) as well as assessment of the model's calibration where this is commonly missing from the literature except for these few studies (Mohsen et al., 2023; Masood et al., 2026). Evaluation in the system level uses the process level usability data generated using the workflow decision points described in Section 3.9.2, where the usefulness of the dialogue is also a factor and clinician-override rates are included in the evaluation directly, which has not appeared in the diabetes-specific XAI literature (Doshi-Velez and Kim, 2017; Abbas, Jeong and Lee, 2025). Moreover, this dual model-level and system-level evaluation structure implements the conclusions that a technically functioning explainable CDSS is not necessarily a clinically effective one, and that this needs to be directly evaluated, reached during the synthesis (Du et al., 2022).

3.12 Ethical Considerations

The study's methodology distinguishively includes data-governance provisions, departing from preceding research that has no explicit data-governance provisions in the reviewed literature, as highlighted in Section 2.6 (Adua et al., 2021; Dada, Birma and Gora, 2024; Ooko and Oginga, 2025). Any covering informed consent procedures for use of patient data for training or validation of the model, data minimisation (section 3.8.1) restricting the type of data captured to that with a clearly established evidentiary basis, and access-controlled secure storage of training data and any prospectively collected clinician interaction data (section 3.9.2). The Diabetes CDSS provisions were selected based on findings from the examined literature on other informatics systems, specifically general data-protection principles, as no diabetes-specific exemplar was found that would allow for the easy adoption of a diabetes-specific governance. Since no evidence of an existing diabetes-specific exemplar of a CDSS was found

that could be directly applied, the provisions considered here are the result of a new contribution that the present study needed to make to the field of diabetes, and as such, informed by general provisions found in the literature around data-protection.

3.13 Chapter Summary

In this chapter, a two-part methodology was presented: One is a systematic literature review guided by the PRISMA checklist, which reduced the number of initially identified studies to 32 studies that were synthesised in Chapter 2, and the other is a design-science-informed conceptual system design, where the identified gaps were translated into a four-layer CDSS architecture, the levelled data-flow design, and the "explainability-integrated clinical workflow. A principle of design science is that when creating a new artefact, its specification should be explicitly traced back to a corresponding analytical finding made in Chapter 2, which is based on a valid knowledge base (Hevner et al, 2004; Peffers et al, 2007). The following chapter discusses the practical implementation of the design, the specific data sets used, algorithms implemented, and the evaluation results when developing the model.

Appendix A: Literature Tracking Table

The table below summarises all 32 studies retained after the PRISMA screening process (Section 3.3.6), recording the extraction fields defined in Section 3.3.8: geographic/population context, method or algorithm, dataset, explainability treatment, key focus, and principal limitation.

No.	Author(s) & Year	Context/Country	Method/Algorithm	Dataset	Explainability	Key Focus	Limitation
1	Abbas, Jeong and Lee (2025)	Cross-domain (CDSS)	N/A (meta-analysis)	62 peer-reviewed XAI-CDSS studies	Cross-method review (SHAP/LIME/Grad-CAM)	Meta-analysis of XAI methods and usability across CDSS domains	Usability evaluation found to be least developed component
2	Adda (2025)	Africa (AWI-Gen, multi-country)	Comparative ML classifiers	AWI-Gen cohort (balanced)	No	Comparative evaluation of classifiers for T2D prediction	Focus on performance metrics only; no explainability
3	Adua et al. (2021)	Ghana	Feature-importance ML model	Local Ghanaian cohort	Implicit (feature importance)	Early detection of T2DM with feature-importance analysis	Single-site design; small sample size
4	American Diabetes Association (2022)	Global (standards)	N/A (clinical guideline)	N/A	No	Diagnostic and classification standards for diabetes	Not an empirical predictive-modelling study
5	Anteneh et al. (2025)	Ethiopia	Gradient boosting and ML models	Ethiopian Public Health Institute data (n>9,000)	No	Predicting CVD among diabetic patients	Comorbidity-focused, not individual diabetes risk screening
6	Berner (2009)	Global (CDSS)	N/A (state-of-the-art report)	N/A	No	Foundational CDSS design requirements (AHRQ)	Pre-dates AI-specific CDSS developments
7	Chari et al. (2023)	Global	Post-hoc explanation contextualisation	T2D risk model dataset	Yes (contextualised post-hoc)	Contextualising explanations for clinical assessment	Presentation-focused; limited deployment testing
8	Choi et al. (2020)	South Korea	ML (multiple classifiers)	Cardiovascular-risk cohort, 5-yr follow-up	No	New-onset diabetes prediction in CV-risk patients	Demographically homogeneous population
9	Dada, Birma and Gora (2024)	Nigeria (Maiduguri)	Ensemble ML	Local clinic EHR data	No	Ensemble-based diabetes prediction	Lifestyle variables unaccounted for; single-site
10	Doshi-Velez and Kim (2017)	N/A (methodological)	N/A (conceptual framework)	N/A	Conceptual (interpretability taxonomy)	Rigorous evaluation framework for interpretable ML	Conceptual only; no empirical application
11	Du et al. (2022)	High-income setting (Ireland)	ML-based CDSS	Gestational diabetes clinical dataset	Yes (SHAP)	Explainable CDSS for gestational diabetes	Validated only in high-resource infrastructure setting
12	Garg et al. (2005)	Global (CDSS)	N/A (systematic review)	100 CDSS trials	No	Effects of CDSS on practitioner performance/outcomes	Outcome-level evidence weaker than process-level evidence
13	Gatimu, Milimo and Sebastian (2016)	Ghana	N/A (epidemiological)	National survey of older adults	No	Prevalence/determinants of diabetes in older Ghanaians	Epidemiological only; not predictive modelling
14	Grundlingh et al. (2022)	South Africa	N/A (epidemiological)	National health survey data	No	Prevalence and risk factors of diabetes/pre-diabetes	Regional South African context; not Ghana-specific
15	International Diabetes Federation (2021)	Global	N/A (reference atlas)	Global surveillance data	No	Global/regional diabetes prevalence statistics	Reference report; not primary research
16	Kawamoto et al. (2005)	Global (CDSS)	N/A (systematic review)	70 CDSS trials	No	Features critical to CDSS trial success	Predates AI/ML-based CDSS
17	Kelkay et al. (2026)	Ethiopia	Deep learning (forecasting)	National diabetes burden data	No	Forecasting regional diabetes trends	Population-level forecasting, not individual prediction
18	Khokhar et al. (2025)	Global	XGBoost + SHAP (SMOTE-balanced)	Public benchmark dataset	Yes (SHAP)	Transparent and accurate T2D prediction	Functionally-grounded evaluation only
19	Lundberg and Lee (2017)	N/A (methodological)	SHAP framework	N/A	Yes (SHAP origin paper)	Unified game-theoretic approach to model explanation	Foundational method; no clinical application
20	Maimaitijiang, Aihaiti and Mamatjan (2025)	Global	Explainable ML + chatbot	Online diabetes risk dataset	Yes (chatbot-mediated)	Patient-facing explainable risk prediction	Patient- not clinician-facing explanation design
21	Majyambere et al. (2026)	Global	Two-stage explainable ML	Screening-stage dataset	Yes	Two-stage explainable T2D risk screening	Not validated in low-resource clinical settings
22	Masood et al. (2026)	Global	N/A (systematic review)	AI-diabetes literature corpus	Partial (reviewed as theme)	AI-driven approaches to diabetes prediction	Notes benchmark-dataset dependency across field
23	Mohsen et al. (2023)	Global	N/A (scoping review)	AI-diabetes literature corpus	Partial (reviewed as theme)	Scoping review of AI methods for diabetes risk	Finds internal-validation-only reporting dominant
24	Muhammad, Algehyne and	Global	Supervised ML (multiple	Public benchmark dataset	No	Predictive supervised ML models for	Ensemble/tree methods

Chapter 3: Methodology — Appendix

No.	Author(s) & Year	Context/Country	Method/Algorithm	Dataset	Explainability	Key Focus	Limitation
	Usman (2020)		algorithms)			diabetes	competitive with complex models
25	Olorunfemi et al. (2025)	Global	Improved ensemble method	Public benchmark dataset	No	Efficient diabetes diagnosis via ensemble method	High accuracy without deep architectures
26	Ooko and Oginga (2025)	Africa (clinic-based)	ML (edge-deployable)	African clinic dataset (n=768)	No	Early detection of chronic disease, deploy ability focus	Small sample; single clinic source
27	Rafie et al. (2025)	Global	XGBoost + explainable AI	Public benchmark dataset	Yes (SHAP-based)	Accurate and explainable T2D prediction	Aggregated feature importance, not patient-specific
28	Rhee et al. (2021)	South Korea	Deep learning	Nationwide population cohort	No	DL-based diabetes prediction, national cohort	Marginal gains over classical ML unclear
29	Ribeiro, Singh and Guestrin (2016)	N/A (methodological)	LIME framework	N/A	Yes (LIME origin paper)	Model-agnostic local explanation method	Sensitive to sampling procedure; noted instability
30	Sutton et al. (2020)	Global (CDSS)	N/A (overview paper)	N/A	No	CDSS benefits, risks and strategies for success	Identifies alert fatigue/automation bias risks
31	Ukoba et al. (2025)	Nigeria	N/A (systematic review)	Nigerian ML-diabetes literature	Partial (reviewed as theme)	Systematic review of ML methods in Nigeria	Notes inconsistent metric reporting across studies
32	Wright et al. (2009)	USA (commercial systems)	N/A (survey)	Commercial clinical information systems	No	CDSS capabilities of commercial systems	Documents historical under-delivery on CDSS features